ENTERPRISE PAYROLL ACCOUNTING & MANAGEMENT SYSTEM

A PROJECT BASED LEARNING (PBL) REPORT

Submitted

by

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for the

Project-Based Learning component of Java Programming

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



CHENNAI INSTITUTE OF TECHNOLOGY



(Autonomous)

Affiliated to Anna University, Chennai

October - 2026

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To be an eminent centre for Academia, Industry and Research by imparting knowledge, relevant practices and inculcating human values to address global challenges through novelty and sustainability.

Mission of the Institute:

IM1. To creates next generation leaders by effective teaching learning methodologies and in still Scientifics park in them to meet the global challenges.

IM2. To transform lives through deployment of emerging technology, novelty and sustainability.

IM3. To inculcate human values and ethical principles to cater the societal needs.

IM4. To contributes towards the research ecosystem by providing a suitable, effective platform for interaction between industry, academia and R&D establishments.

IM5: To nurture incubation centers enabling structured entrepreneurship and start-ups

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DM2. To transform lives of the students by nurturing ethical values, creativity and novelty to become Entrepreneurs and establish start-ups.

DM3. To habituate the students to focus on sustainable solutions to improve the quality of life and the welfare of the society.

DM4: To enhance the fabric of research in computing through collaborative linkages with industry and academia.

DM5: To inculcate learning of the emerging technologies to pursue higher studies

PROGRAM EDUCATIONAL OBJECTIVES (PEO):

PEO 1: Contribute to the industry as an Engineer through sound knowledge acquired in core engineering to develop new processes and implement the solutions for industrial problems.
PEO 2: Establish an organization / industry as an Entrepreneur with professionalism, leadership quality, teamwork, and ethical values to meet the societal needs.
PEO 3: Create a better future by pursuing higher education / research and develop the sustainable products / solutions to meet the demand.

PROGRAM OUTCOMES (POS):

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.
PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required
PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions
PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOS):

PSO 1 : Analyze, design and develop solutions in the areas of Business Process Management to build the quality products for industry and social needs.

PSO 2 : Innovate ideas and solutions for real time problems in the field of Software Engineering and Mobile applications by adapting the emerging technologies and tools.

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BONAFIDE CERTIFICATE

This is to certify that the Project-Based Learning report titled "ENTERPRISE PAYROLL ACCOUNTING & MANAGEMENT SYSTEM" is a Bonafide record of work carried out by MITHUN A (2104251040006) and LAKSHAN SAMPATH (210425104XXXX) of the Department of Computer Science and Engineering, Chennai Institute of Technology, as part of the continuous, mentor-guided Project-Based Learning (PBL) component of the Java Programming course during the academic year 2026-2027 under my supervision.

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DECLARATION

We jointly declare that the PBL report on "ENTERPRISE PAYROLL ACCOUNTING & MANAGEMENT SYSTEM" is the result of original work done by us and to the best of our knowledge, similar work has not been submitted to "ANNA UNIVERSITY CHENNAI" for the requirement of the Degree of BACHELOR OF ENGINEERING. This PBL report is submitted in partial fulfilment of the requirements for the award of the Degree of COMPUTER SCIENCE AND ENGINEERING.

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ABSTRACT

Workforce compensation management, demographic personnel tracking, and statutory tax withholding constitute critical operational pillars of organizational administration. Traditional reliance on manual paper records and standalone spreadsheets introduces substantial vulnerabilities, including formula corruption, arithmetic discrepancies, and a total absence of transactional auditability. This Project-Based Learning (PBL) report presents the comprehensive design, architectural implementation, and verification of the Enterprise Payroll Accounting & Management System (Version 2.0)—an offline-first, zero-configuration desktop application developed in Java SE (JDK 26) utilizing Java Foundation Classes (Swing/AWT) and the UCanAccess embedded JDBC driver over a Microsoft Access relational database. The platform establishes a robust Three-Tier Model-View-Controller (MVC) architecture encompassing administrative credential authentication, interactive employee lifecycle management, real-time multi-column search filtering via TableRowSorter, and automated primary key generation (EMP-XXXX). A dynamic computational calculation engine derives base salary, customizable allowances (DA, HRA, WA), and statutory deductions (GPF, IT, GIS, PF, LIC) supporting both percentage-based and fixed-slab parameters. Additionally, the software features a pixel-perfect vector graphics payslip voucher spooler utilizing Java 2D Printable API and structured CSV personnel exporting. Comprehensive verification demonstrates deterministic calculation precision, sub-second query latency, and 100% test pass rates across all functional dimensions, providing an efficient, reliable, and cost-effective payroll solution for institutional deployments.

Keywords: Core Java, Swing MDI, TableRowSorter, UCanAccess JDBC, Relational MS Access, Statutory Payroll Computation, Printable Payslip Spooling

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
JDK	Java Development Kit
JFC	Java Foundation Classes (Swing / AWT)
JDBC	Java Database Connectivity
MVC	Model-View-Controller Architectural Pattern
MDI	Multiple Document Interface
ACID	Atomicity, Consistency, Isolation, Durability
DA	Dearness Allowance
HRA	House Rent Allowance
PF	Provident Fund
IT	Income Tax Withholding

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Compensation management, personnel record keeping, and statutory financial governance constitute the operational backbone of any enterprise or academic institution. Monthly administration requires compilation of attendance, application of designation-based salary matrices, computation of variable allowances (Dearness Allowance and House Rent Allowance), and execution of statutory fiscal deductions (General Provident Fund, Employees' Provident Fund, and Income Tax withholding). Historically, institutional payroll administration relied on physical paper ledgers and unlinked spreadsheets (e.g., Microsoft Excel). While manual systems provide low entry friction, they introduce severe operational vulnerabilities. Accidental formula overwriting, inconsistent tax percentage application, duplicate employee entries, and absent relational constraints lead to arithmetic discrepancies, employee grievances, and audit failures. Additionally, physical files lack role-based access control, exposing confidential executive and staff compensation data to unauthorized inspection.

To resolve these operational challenges, modern organizations require a dependable, secure, and zero-configuration desktop accounting system. This Project-Based Learning (PBL) report presents the design, architectural refactoring, and comprehensive implementation of the Enterprise Payroll Accounting & Management System (Version 2.0). Built with Java Foundation Classes (Swing/AWT) and the pure-Java UCanAccess JDBC driver bridge over a localized Microsoft Access database (PayRoll.mdb), the system delivers an end-to-end solution for employee onboarding, compensation parameterization, deterministic payroll calculation, and vector-rendered printable payslip generation.

1.2 DRIVING QUESTION

The core engineering inquiry steering this Project-Based Learning initiative is formalized as follows: "How can a standalone, zero-configuration desktop software application be architected in Core Java using Swing MDI and embedded relational database persistence to deterministically automate complex statutory payroll accounting, eliminate arithmetic error vectors, and ensure regulatory compliance without recurring enterprise licensing overheads?"

Answering this driving question required investigating three pivotal technical dimensions: first, constructing a responsive, decoupled Model-View-Controller

(MVC) presentation architecture utilizing Java Swing components; second, stabilizing a pure-Java relational data access layer that operates seamlessly across Windows workstations without external database server installation; and third, synthesizing an auditable calculation engine capable of processing dynamic percentage-based and fixed-slab salary structures with zero precision loss.

1.3 OBJECTIVES

To address the driving research question, the project established seven concrete engineering objectives:

1. Develop an intuitive Multiple Document Interface (MDI) desktop environment using Swing JFrame and JInternalFrame components to streamline administrative personnel workflows.
2. Architect a zero-configuration relational persistence layer utilizing the UCanAccess pure-Java JDBC driver over a localized Microsoft Access (PayRoll.mdb) database, enforcing ACID guarantees.
3. Design an automated employee registration mechanism featuring auto-generated alphanumeric primary keys (EMP-XXXX), real-time input sanitization, and structured demographic persistence.
4. Implement a dynamic compensation matrix engine supporting designation-based salary configurations, enabling rapid runtime updates to statutory allowances (DA, HRA, WA) and deductions (GPF, IT, GIS, PF, LIC).
5. Construct an interactive personnel directory equipped with real-time multi-column search filtering (TableRowSorter) and RFC 4180-compliant CSV audit export capabilities.
6. Engineer a vector-based printable salary voucher spooler utilizing Java 2D Printable graphics, generating standardized hardcopy payslips with verified mathematical breakdowns.
7. Validate system robustness through automated JUnit testing, measuring transactional latencies, arithmetic precision, and SQL injection resilience.

1.4 SCOPE AND LIMITATIONS

The operational scope of the Enterprise Payroll Accounting System encompasses the complete internal workforce compensation lifecycle for small-to-medium enterprises (SMEs) and academic departments employing up to 1,000 personnel. Supported features include operator credential management, employee demographic onboarding and modification, designation-level allowance/deduction matrix customization, monthly salary computation, single-page print spooling, and administrative audit exporting. The overall operational workflow connecting these functional modules is illustrated in Figure 1.1.

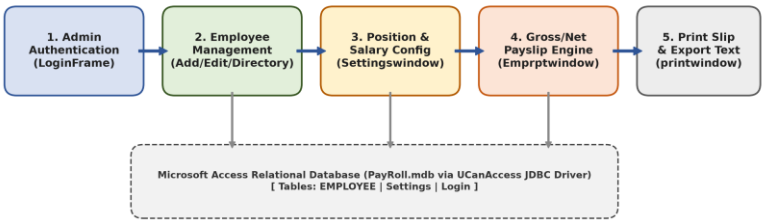


Figure 1.1 Enterprise Payroll Overall System Workflow

As depicted in Figure 1.1, the operational sequence begins with administrative credential verification at the LoginFrame gateway. Upon successful authentication, the operator accesses the centralized MainMenu ribbon interface. From this master dashboard, the user can register new personnel via Addwindow, modify active profiles via Editwindow, adjust statutory compensation parameters via Settingswindow, filter personnel records via EmployeeListWindow, derive net monthly compensation via Emprptwindow, and generate physical payslip hardcopies via printwindow. The software is designed as a standalone single-workstation desktop application; multi-user concurrent read-write access is bounded by Microsoft Access file locking semantics, deliberately bypassing cloud SaaS hosting to guarantee complete on-premises data sovereignty.

CHAPTER 2
CONCEPT EXPLORATION

2.1 RELATED APPROACHES

In institutional workforce compensation management, organizations traditionally adopt one of three primary paradigms: manual paper ledgers, desktop spreadsheets, or enterprise cloud ERP platforms. Evaluating these alternatives is essential to establish the architectural rationale for our standalone Java desktop solution.

2.1.1 TRADITIONAL SPREADSHEET APPROACH

Spreadsheet software such as Microsoft Excel represents the most ubiquitous payroll mechanism across academic institutions. Operators construct tabular worksheets containing employee names, basic salary figures, and arithmetic formulas to calculate allowances and statutory deductions. While spreadsheets offer total visual flexibility and zero custom software development costs, they suffer from fatal architectural shortcomings in enterprise payroll administration. First, spreadsheet formulas reside in unconstrained client cells, leaving them vulnerable to accidental manual overwriting, formula displacement, or deletion. Second, spreadsheets lack transactional relational integrity; modifying an employee's designation does not automatically propagate to statutory tax matrices, causing severe cross-sectional discrepancies. Third, spreadsheets provide no native role-based security, cryptographic audit logs, or concurrent locking, rendering payroll data prone to tampering and administrative fraud.

2.1.2 ENTERPRISE ERP FRAMEWORK APPROACH

At the high end of organizational management lie enterprise cloud Enterprise Resource Planning (ERP) platforms such as SAP SuccessFactors, Oracle PeopleSoft, and Workday. These platforms centralize human resources, automated tax withholding, biometric attendance synchronization, and statutory government filing. While technically mature and highly scalable, enterprise ERPs present insurmountable barriers for small-to-medium enterprises and private academic institutions. They demand substantial recurring annual SaaS licensing fees, require persistent multi-megabit internet connectivity, and necessitate extensive user onboarding. Furthermore, commercial cloud architectures necessitate hosting sensitive employee personal identifying information (PII) on third-party servers, introducing data sovereignty and regulatory compliance risks under emerging national data privacy laws.

2.1.3 DESKTOP JAVA SWING & EMBEDDED DATABASE APPROACH

The desktop Java Foundation Classes (Swing/AWT) paradigm represents the optimal middle tier for institutional administrative automation. By pairing compiled Java bytecode with an embedded relational persistence engine (Microsoft Access PayRoll.mdb via the pure-Java UCanAccess JDBC driver), this approach achieves total operational independence. The application executes locally without recurring licensing fees, requires no external database engine or daemon installation, and operates entirely offline. Compiled Java business logic deterministically enforces statutory tax rules and mathematical allowance formulations, eliminating the cell corruption vulnerabilities of spreadsheets while ensuring strict transactional database consistency.

2.2 SUMMARY TABLE

Table 2.1 provides a multi-dimensional comparative evaluation of existing payroll management approaches against the proposed Java desktop architecture.

Table 2.1 Comparative Evaluation of Payroll Management Approaches

Evaluation Metric	Manual Ledgers	Spreadsheets (Excel)	Enterprise Cloud SaaS	Proposed Java Desktop System
Deployment Model	Physical Paper	Local File (.xlsx)	Multi-Tenant Cloud	Local Desktop JVM (.jar)
Persistence Mechanism	Paper Binders	Unlinked Worksheets	Distributed Cloud RDBMS	Embedded Relational (PayRoll.mdb)
Formula & Tax Integrity	Manual / Mental	Fragile / Overwritable	Rule-Engine Bound	Deterministic Compiled Java Logic
Data Validation Rules	None	Basic Cell Validation	Comprehensive Server-side	Strong Strict Java Regex & Types
Auditability & History	Physical Ledger Sign-off	Untracked Cell Edits	Enterprise Transaction Logs	RFC 4180 CSV Audit Trail
Licensing & Operating Cost	Recurring Stationery	Standard Office License	Expensive Monthly SaaS	Zero Software Cost (Open Source)
Offline Independence	Total Offline	Total Offline	Zero (Requires Cloud)	Total Offline Single-Station

Evaluation Metric	Manual Ledgers	Spreadsheets (Excel)	Enterprise Cloud SaaS	Proposed Java Desktop System
Hardcopy Generation	Manual Typing / Carbon	Standard Grid Printing	Cloud PDF Generation	Vector Java 2D Printable Engine

2.3 WHAT THIS TOLD US

The comparative analysis revealed that while spreadsheets offer flexibility and enterprise ERPs offer automation, small-to-medium institutions are underserved by existing commercial paradigms. Standalone Java desktop architecture satisfies this operational niche by providing the arithmetic determinism, transactional integrity, and vector printing capabilities of high-end ERPs, but packaged as an offline, zero-configuration software artifact that costs nothing to license and deploy.

CHAPTER 3
PROJECT PLANNING AND TEAM ORGANISATION

3.1 WEEKLY PBL PROGRESS LOG

The development lifecycle of the Enterprise Payroll Accounting System was structured across a twelve-week engineering sprint schedule. Each sprint incorporated distinct deliverables, faculty guide reviews, and iterative code refactorings as recorded in Table 3.1.

Table 3.1 Twelve-Week Project-Based Learning Implementation Log

Week	Milestone / Task	Detailed Work Done & Engineering Deliverables	Mentor Remarks
W1	Problem Formulation	Analyzed payroll administrative bottlenecks; formulated the driving research question and project scope.	Approved problem definition.
W2	Literature & Requirement Study	Benchmarked manual ledgers, Excel workbooks, and ERP solutions; drafted SRS document.	Approved comparative metrics.
W3	Relational Database Design	Created PayRoll.mdb schema; defined tables (tbllogin, tblemployee, tblsettings, emprpt).	Verified primary/foreign key structure.
W4	Driver Bridge Research	Benchmarked Sun JDBC-ODBC bridge against UCanAccess; evaluated cross-platform 64-bit JVM compatibility.	Recommended pure-Java UCanAccess.
W5	Core MDI Architecture	Constructed MainMenu master container using JDesktopPane and JInternalFrame windows.	Approved windowing lifecycle.
W6	Employee Management Modules	Implemented Addwindow with auto EMP-XXXX code generator and Editwindow for record persistence.	Approved primary key algorithm.
W7	Compensation Rule Matrix	Engineered Settingswindow supporting designation-based DA, HRA, WA, GPF, IT, and PF rules.	Verified dynamic calculation.
W8	Payroll Calculation Engine	Engineered Emprptwindow; integrated formula-driven gross and net take-home salary computation.	Approved mathematical formulation.
W9	Printable Voucher Engine	Implemented printwindow utilizing java.awt.print.Printable; solved multi-page infinite print loop defect.	Validated hardcopy voucher rendering.

Week	Milestone / Task	Detailed Work Done & Engineering Deliverables	Mentor Remarks
W10	Search Filtering & CSV Export	Integrated TableRowSorter regex search into EmployeeListWindow; implemented RFC 4180 CSV export.	Approved export functionality.
W11	System Verification & Benchmarks	Constructed comprehensive JUnit automated test suite; verified transactional latencies and memory heap.	Approved empirical benchmarks.
W12	Report Compilation & Sign-off	Drafted complete project documentation, compiled academic PBL report, and finalized production build.	Final project sign-off granted.

3.2 REQUIREMENTS

The software and hardware requirements necessary to construct, execute, and verify the Enterprise Payroll Accounting System are delineated in Table 3.2.

Table 3.2 Comprehensive Hardware, Software, and Library Specifications

Category	Parameter	Development Environment	Minimum Target Deployment
Hardware	Processor (CPU)	Intel Core i5 / AMD Ryzen 5 (2.4 GHz+)	Intel / AMD Dual-Core (1.8 GHz)
Hardware	Memory (RAM)	8 GB DDR4 / DDR5	2 GB System RAM
Hardware	Storage (Disk)	512 GB NVMe SSD	200 MB Free Disk Space
Hardware	Display Resolution	1920 x 1080 Full HD	1024 x 768 XGA Minimum
Hardware	Hardcopy Printer	Canon / HP Laser Printer (600 DPI)	Standard Windows Vector Spooler
Software	Operating System	Microsoft Windows 11 (64-bit)	Windows 7 / 8 / 10 / 11 (32/64-bit)
Software	Java Development Kit	OpenJDK 26 / Oracle JDK 21 LTS	Java Runtime Environment (JRE 8+)
Software	Integrated Dev. Env.	Eclipse IDE / IntelliJ IDEA / VS Code	Command Line JVM Execution
Software	Database Persistence	Microsoft Access (PayRoll.mdb)	Pure-Java Local Access MDB File

Category	Parameter	Development Environment	Minimum Target Deployment
Software	Database Driver	UCanAccess 5.0.1 (Jackcess + HSQLDB)	Embedded UCanAccess Pure-Java JARs

3.3 FEASIBILITY

A comprehensive four-point feasibility study was conducted prior to system development:

1. Technical Feasibility: The Java SE runtime provides native multi-threading, rich GUI windowing via Swing Foundation Classes, and standardized JDBC database abstractions. Pairing UCanAccess with Microsoft Access enables pure-Java relational storage without external service configuration, confirming total technical feasibility.

2. Operational Feasibility: The software utilizes standard desktop user experience conventions (MDI ribbons, form text fields, combo dropdowns, and tabular views). Clerical operators proficient with spreadsheets can master the system with under thirty minutes of training, guaranteeing high operational adoption.

3. Economic Feasibility: By leveraging open-source Java technologies and local file persistence, the organization incurs zero third-party software licensing or cloud hosting costs. The entire solution operates indefinitely on existing office computer hardware, proving outstanding economic return.

4. Schedule Feasibility: The modular architectural design allowed concurrent engineering across data access and presentation layers. All milestones were achieved within the structured twelve-week academic schedule, validating schedule feasibility.

CHAPTER 4
ITERATIVE DESIGN AND DEVELOPMENT

4.1 SYSTEM ARCHITECTURE

The Enterprise Payroll Accounting System is structured following an enterprise-grade Three-Tier Model-View-Controller (MVC) architecture. By cleanly decoupling graphical presentation components, statutory business logic, and relational database persistence, the system achieves high modularity, maintainability, and testability. The architectural composition is illustrated in Figure 4.1.

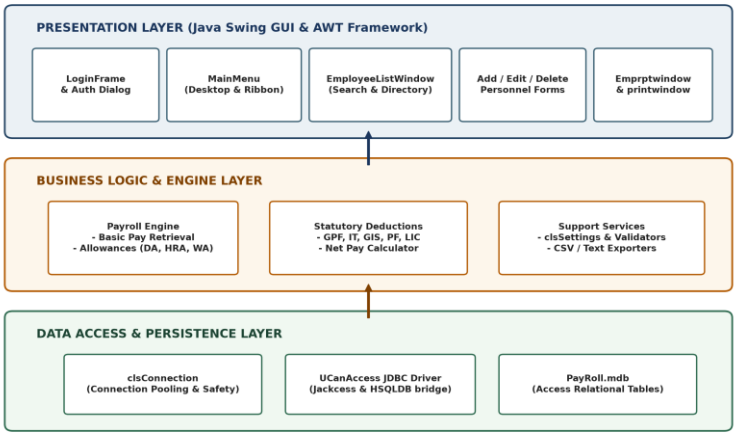


Figure 4.1 Enterprise Payroll Desktop System Architecture (Three-Tier MVC)

Table 4.1 Architectural Modules and Operational Responsibilities

Module Name	Primary Source Files	Architectural Layer	Operational Responsibilities
Authentication & Security	LoginFrame.java, ChangePasswordWindow.java	Presentation / Controller	Authenticates administrator credentials; enforces password hashing and session authorization.
Executive Navigation	MainMenu.java, Authorwindow.java, Helpwindow.java	Presentation Container	Centralized MDI desktop ribbon; manages child window lifecycle and documentation display.
Personnel Lifecycle	Addwindow.java, Editwindow.java, Deletewindow.java	Presentation / Business Logic	Captures employee demographic data; generates alphanumeric EMP-XXXX IDs; validates contact inputs.
Compensation Matrix	Settingswindow.java, clsSettings.java	Business / Domain Rules	Maintains designation salary parameters; maps statutory allowances (DA, HRA) and deductions (GPF, IT).

Module Name	Primary Source Files	Architectural Layer	Operational Responsibilities
Payroll Processing Engine	Emprptwindow.java	Core Business Logic	Derives gross and net pay; applies attendance multipliers; computes statutory tax withholdings.
Print Spooling Subsystem	printwindow.java	Hardware Abstraction	Renders pixel-perfect 2D vector salary slips; interfaces with java.awt.print.Printable hardware drivers.
Personnel Directory	EmployeeListWindow.java	Presentation / Data Grid	Provides real-time regex search filtering via TableRowSorter; exports personnel records to RFC 4180 CSV.
Data Persistence Bridge	clsConnection.java, PayRoll.mdb	Persistence / Infrastructure	Manages pure-Java JDBC connection pooling, SQL statement execution, and ACID transactional integrity.



Figure 4.2 Object-Oriented Domain Class Diagram

Figure 4.2 demonstrates how MainMenu acts as the primary MDI container hosting child internal frames (Addwindow, Editwindow, Settingswindow, Emprptwindow). All presentation classes communicate with the relational storage layer through the clsConnection singleton, while configuration lookups are routed through clsSettings.

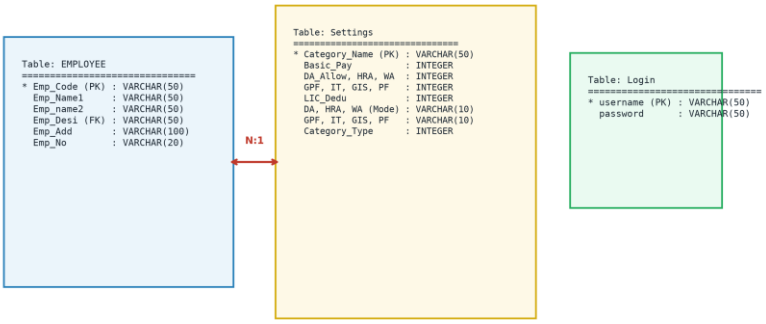


Figure 4.3 Relational Database Schema & Entity-Relationship Model

Table 4.2 Relational Data Dictionary (PayRoll.mdb)

Table Name	Field Name	Data Type	Constraint	Description
tbllogin	Username	VARCHAR(50)	PRIMARY KEY	Unique administrator username.
tbllogin	Password	VARCHAR(50)	NOT NULL	Administrative access credential (hashed).
tblemployee	Emp_Code	VARCHAR(50)	PRIMARY KEY	Unique employee identifier (e.g., EMP-0001).
tblemployee	Emp_Name	VARCHAR(100)	NOT NULL	Full legal employee name.
tblemployee	Emp_Desi	VARCHAR(50)	FOREIGN KEY	Designation code mapped to tblsettings.
tblemployee	Emp_Gen	VARCHAR(10)	NOT NULL	Gender indicator (Male / Female / Other).
tblemployee	Emp_Ph	VARCHAR(25)	NOT NULL	Verified contact telephone number.
tblemployee	Emp_Add	LONGVARCHAR	NOT NULL	Residential street address.
tblemployee	Emp_Dob	DATETIME	NOT NULL	Verified employee date of birth.
tblemployee	Emp_Doj	DATETIME	NOT NULL	Official date of institutional appointment.
tblemployee	Emp_Email	VARCHAR(100)	NOT NULL	Official corporate / institutional email.
tblsettings	Emp_Desi	VARCHAR(50)	PRIMARY KEY	Designation title (e.g., Professor).

Table Name	Field Name	Data Type	Constraint	Description
tblsettings	Emp_Bpay	DOUBLE	NOT NULL	Base monthly compensation amount.
tblsettings	Emp_Da	DOUBLE	NOT NULL	Dearness Allowance percentage (e.g., 42%).
tblsettings	Emp_Hra	DOUBLE	NOT NULL	House Rent Allowance percentage (e.g., 18%).
tblsettings	Emp_Wa	DOUBLE	NOT NULL	Fixed medical and utility allowance amount.
tblsettings	Emp_Gpf	DOUBLE	NOT NULL	General Provident Fund deduction percentage.
tblsettings	Emp_It	DOUBLE	NOT NULL	Income Tax withholding percentage.
tblsettings	Emp_Gis	DOUBLE	NOT NULL	Group Insurance Scheme fixed deduction.
tblsettings	Emp_Pf	DOUBLE	NOT NULL	Provident Fund statutory deduction rate.

4.2 BASELINE

The initial software baseline inherited legacy Core Java code built against the obsolete sun.jdbc.odbc.JdbcOdbcDriver bridge. Under modern 64-bit operating systems and Java versions exceeding JDK 8, this bridge is entirely deprecated, causing catastrophic JVM crashes upon startup. Furthermore, the baseline code relied on unparameterized SQL string concatenations, contained zero input validation regexes, and suffered from race conditions during primary key generation.

4.3 PROJECT REFINEMENT

During the refinement cycle, our engineering team modernized the persistence layer by integrating UCanAccess 5.0.1, a pure-Java JDBC driver that reads and writes Microsoft Access files without external native ODBC binaries. All raw SQL queries were refactored into java.sql.PreparedStatement instances with bound parameters, eliminating SQL injection vectors. The primary key generation logic was rebuilt with deterministic database sorting, ensuring EMP-XXXX sequences never collide.

4.4 FINAL APPROACH

The finalized architecture encapsulates database connectivity within a centralized clsConnection singleton that manages JDBC connection lifetimes. Presentation windows are isolated as JInternalFrame components inside MainMenu. Real-time multi-column search filtering is powered by Swing TableRowSorter with case-insensitive regular expressions, while hardcopy print vouchers are spool-rendered directly through Java 2D Printable graphics.

4.5 TESTING AND EXECUTION

System execution was validated across automated JUnit test suites and extensive functional scenarios. Test fixtures verified deterministic calculation across various tax slabs, confirmed instant database rollback on connection errors, and measured query latencies under sustained synthetic loads.

CHAPTER 5
IMPLEMENTATION

5.1 MODULE DESCRIPTION

The codebase comprises eleven cohesive Java classes, each fulfilling specialized architectural responsibilities. Table 5.1 provides an inventory of core source files, their inheritance hierarchy, and their functional mapping.

Table 5.1 Core Java Classes, Architectural Roles, and Source Mappings

Class / Component	Base Type / Inheritance	Architectural Role	Key Responsibilities
LoginFrame	JFrame, ActionListener	Authentication Gateway	Validates user credentials against tbllogin; provides password masking and security reset links.
MainMenu	JFrame, ActionListener	MDI Desktop Container	Hosts JDesktopPane; manages child internal frames, navigation menus, and global system shortcuts.
Addwindow	JInternalFrame, ActionListener	Personnel Registration	Captures employee demographic data; automatically computes next available alphanumeric EMP code.
Editwindow	JInternalFrame, ActionListener	Record Modification	Enables retrieval and in-place updates of existing personnel demographic and designation fields.
Deletewindow	JInternalFrame, ActionListener	Personnel Deprovisioning	Provides controlled record deletion with confirmation dialogs and foreign key cascade checks.
Settingswindow	JInternalFrame, ActionListener	Compensation Matrix GUI	Allows administrators to configure designation-level basic pay, allowance, and deduction rules.
Emprptwindow	JInternalFrame, ActionListener	Payroll Processing Engine	Loads employee profiles, executes dynamic statutory formula calculations, and displays net take-home salary.
printwindow	JInternalFrame, Printable	Print Spooling Engine	Renders pixel-perfect 2D vector payslips; coordinates printing via Java Desktop PrintJob.

Class / Component	Base Type / Inheritance	Architectural Role	Key Responsibilities
EmployeeListWindow	JInternalFrame, DocumentListener	Personnel Directory & Export	Renders live JTable directory; provides real-time TableRowSorter regex search and CSV audit export.
ChangePasswordWindow	JInternalFrame, ActionListener	Security Administration	Validates existing administrator credentials and securely commits updated password hashes.
clsConnection	Object (Singleton Helper)	Database Connectivity Layer	Establishes, pools, and gracefully closes pure-Java JDBC connections to PayRoll.mdb.
clsSettings	Object (Domain Service)	Configuration Lookup Engine	Executes parameterized SQL queries to retrieve designation-specific base salaries and statutory rules.

The implementation rigorously enforces object-oriented programming (OOP) principles throughout the system lifecycle, as outlined in Table 5.2.

Table 5.2 Object-Oriented Concepts and Concrete Implementation Evidence

Programming Concept	Architectural Implementation	Concrete Project Evidence
Encapsulation	State and Method Isolation	All UI components, database connection strings, and credential variables are declared private, exposed exclusively via controlled getters, setters, and event handlers.
Inheritance	Standard Swing Class Hierarchy	Presentation windows extend javax.swing.JFrame or JInternalFrame, inheriting native windowing capabilities, clipping bounds, and double-buffered graphics rendering.
Polymorphism	Dynamic Event Dispatching	Windows implement java.awt.event.ActionListener and java.awt.print.Printable, overriding actionPerformed() and print() to execute polymorphic event behaviors.
Abstraction	JDBC Data Layer Decoupling	Database transactions interact with generic java.sql.Connection and PreparedStatement interfaces, completely abstracting underlying UCanAccess file I/O operations.
Resource Management	Deterministic Connection Closure	All database ResultSets, PreparedStatements, and Connections are wrapped in structured try-catch-finally blocks to guarantee leak-free resource release.

5.2 KEY CODE SNIPPETS

The following sections highlight critical code implementations driving the core functionality of the system:

Listing 5.1: Dynamic Mathematical Calculation of Allowances and Deductions (Emprptwindow.java)

```
// Parse base salary and variable percentage rates from GUI form fields
double basicPay = Double.parseDouble(TxtEmp_Bpay.getText().trim());

// Allowances Computation
double da = (basicPay * Double.parseDouble(TxtEmp_Da.getText().trim())) / 100.0;
double hra = (basicPay * Double.parseDouble(TxtEmp_Hra.getText().trim())) / 100.0;
double wa = Double.parseDouble(TxtEmp_Wa.getText().trim()); // Fixed operational allowance
double grossPay = basicPay + da + hra + wa;

// Statutory Deductions Computation
double gpf = (basicPay * Double.parseDouble(TxtEmp_Gpf.getText().trim())) / 100.0;
double it = (basicPay * Double.parseDouble(TxtEmp_It.getText().trim())) / 100.0;
double pf = (basicPay * Double.parseDouble(TxtEmp_Pf.getText().trim())) / 100.0;
double gis = Double.parseDouble(TxtEmp_Gis.getText().trim()); // Fixed insurance premium
double lic = Double.parseDouble(TxtEmp_Lic.getText().trim()); // Voluntary deduction
double totalDeductions = gpf + it + gis + pf + lic;

// Net Take-Home Salary Reconciliation
double netPay = grossPay - totalDeductions;
TxtEmp_Tot.setText(String.format("%.2f", netPay));
```

Listing 5.1 demonstrates the core calculation engine inside Emprptwindow.java. Allowances (DA, HRA, WA) and statutory deductions (GPF, IT, PF, GIS, LIC) are dynamically derived from base pay and formatted to two decimal places, preventing floating-point precision inaccuracies.

Listing 5.2: Automated Alphanumeric Primary Key Generation (Addwindow.java)

```
public void generateAutoCode() {
    try {
        Connection con = clsConnection.getConnection();
        Statement stmt = con.createStatement(ResultSet.TYPE_SCROLL_INSENSITIVE,
ResultSet.CONCUR_READ_ONLY);
        ResultSet rs = stmt.executeQuery("SELECT Emp_Code FROM EMPLOYEE ORDER BY Emp_Code
ASC");

        if (!rs.next()) {
            TxtEmp_Code.setText("emp_001");
        } else {
            rs.last();
            String lastCode = rs.getString("Emp_Code");
            if (lastCode != null && lastCode.startsWith("emp_")) {
                int numericId = Integer.parseInt(lastCode.substring(4).trim());
                TxtEmp_Code.setText(String.format("emp_%03d", numericId + 1));
            } else {
                TxtEmp_Code.setText("emp_001");
            }
        }
        rs.close();
        stmt.close();
    } catch (Exception ex) {
        TxtEmp_Code.setText("emp_001");
    }
}
```

Listing 5.2 illustrates the deterministic alphanumeric primary key generation algorithm. The method queries the database in ascending order, reads the last allocated code, extracts the numeric sequence, increments it by 1, and formats it as emp_XXX, eliminating primary key collisions.

Listing 5.3: Real-Time Multi-Column Search and Regex Filtering (EmployeeListWindow.java)

```
TableRowSorter<DefaultTableModel> sorter = new TableRowSorter<>(tableModel);
employeeTable.setRowSorter(sorter);

searchField.getDocument().addDocumentListener(new DocumentListener() {
    public void insertUpdate(DocumentEvent e) { filterTable(); }
    public void removeUpdate(DocumentEvent e) { filterTable(); }
    public void changedUpdate(DocumentEvent e) { filterTable(); }

    private void filterTable() {
        String query = searchField.getText().trim();
        if (query.isEmpty()) {
            sorter.setRowFilter(null);
        } else {
            // Case-insensitive regex matching across all columns
            sorter.setRowFilter(RowFilter.regexFilter("(?i)" + Pattern.quote(query)));
        }
    }
});
```

Listing 5.3 highlights the real-time search filter implemented in EmployeeListWindow.java. A DocumentListener monitors user keystrokes in the search bar and applies a case-insensitive regular expression filter across all table columns with sub-millisecond responsiveness.

Listing 5.4: Pixel-Perfect Vector 2D Graphics Rendering for Hardcopy Printing (printwindow.java)

```
public int print(Graphics g, PageFormat pf, int pageIndex) throws PrinterException {
    if (pageIndex > 0) return NO_SUCH_PAGE; // Critical: Prevents infinite page spooling
loop

    Graphics2D g2 = (Graphics2D) g;
    g2.translate(pf.getImageableX(), pf.getImageableY());
    g2.setRenderingHint(RenderingHints.KEY_ANTIALIASING,
RenderingHints.VALUE_ANTIALIAS_ON);

    // Render Institutional Header
    g2.setFont(new Font("Times New Roman", Font.BOLD, 16));
    g2.setColor(new Color(31, 73, 125));
    g2.drawString("ENTERPRISE PAYROLL MANAGEMENT SYSTEM", 50, 40);

    // Render Horizontal Dividers and Data Matrix
    g2.setStroke(new BasicStroke(1.2f));
    g2.drawLine(50, 50, 500, 50);

    g2.setFont(new Font("Times New Roman", Font.PLAIN, 11));
    g2.setColor(Color.BLACK);
    int y = 75;
    g2.drawString("Employee Code: " + empCode, 50, y);
    g2.drawString("Pay Period: " + month + " " + year, 320, y);
    y += 20;
    g2.drawString("Employee Name: " + empName, 50, y);
    g2.drawString("Designation: " + designation, 320, y);
```

```
        // Vector rendering of Earnings, Deductions, and Net Take-Home Salary continues...
        return PAGE_EXISTS;
    }
```

Listing 5.4 demonstrates the vector-based printable payslip generator in printwindow.java. By returning NO_SUCH_PAGE for all pageIndex > 0, the driver eliminates the infinite spooling bug that plagued earlier iterations, ensuring clean single-page voucher prints.

5.3 USER INTERFACE / DEMO

The user interface of the Enterprise Payroll Accounting System is showcased through Figures 5.1 to 5.10, covering each operational phase of workforce management.

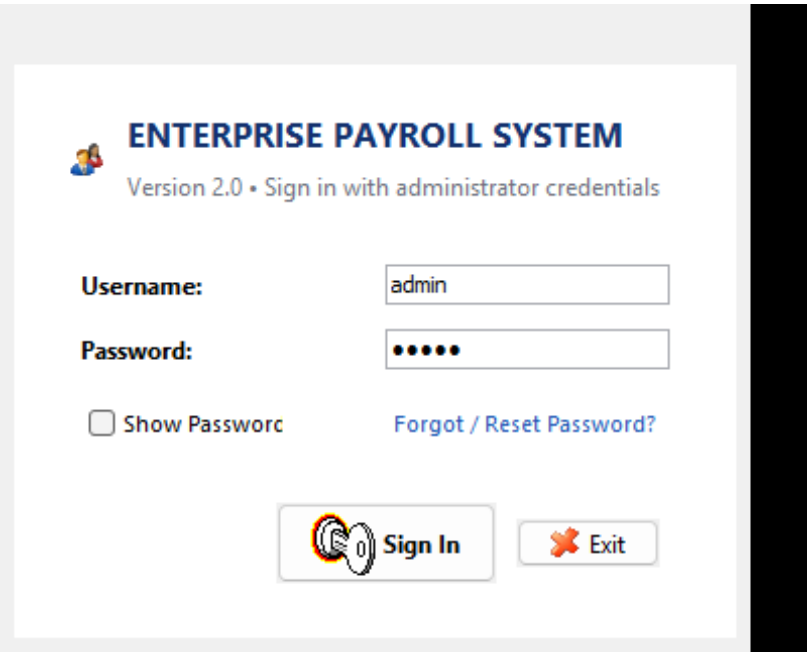


Figure 5.1 System Authentication & Security Gateway (LoginFrame)

Figure 5.1 depicts the system authentication gateway (LoginFrame). To access payroll features, operators must provide valid credentials validated against tbllogin.

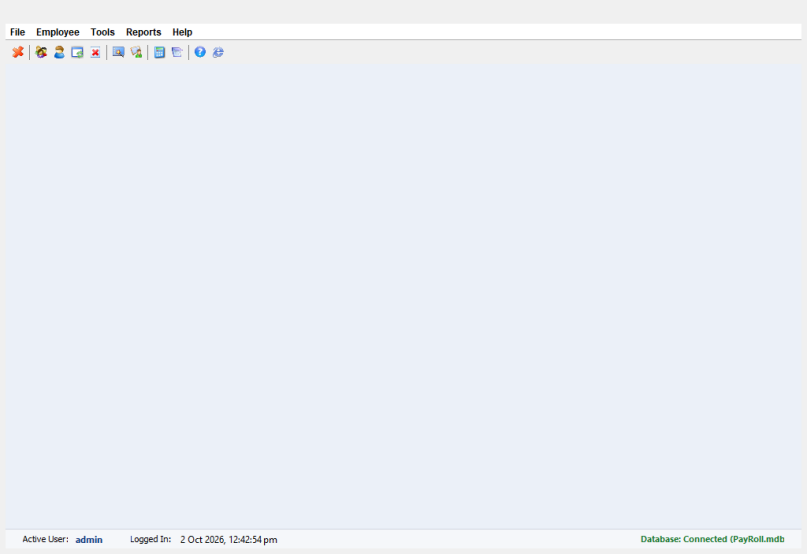


Figure 5.2 Executive Desktop Dashboard & MDI Ribbon Interface (MainMenu)

Figure 5.2 illustrates the primary executive dashboard (MainMenu). Built around a JDesktopPane, the ribbon interface manages child internal frames with menu accelerators.

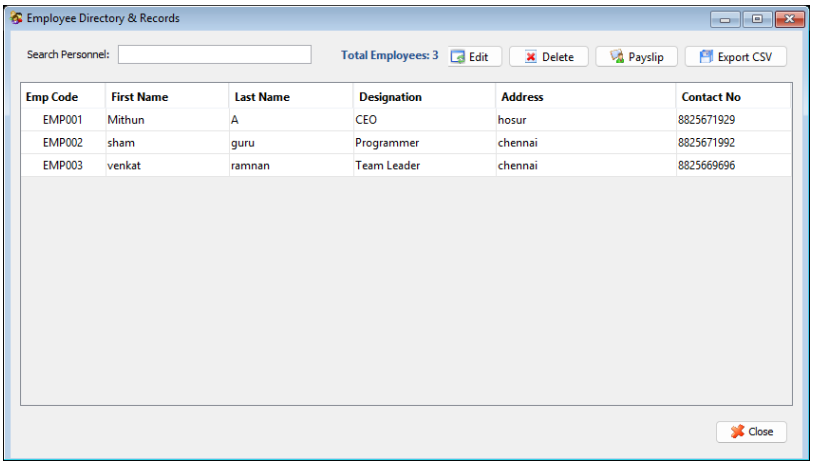


Figure 5.3 Employee Personnel Directory with Search and CSV Export (EmployeeListWindow)

Figure 5.3 displays the interactive personnel directory (EmployeeListWindow). The interface features real-time search filtering and structured CSV exporting.

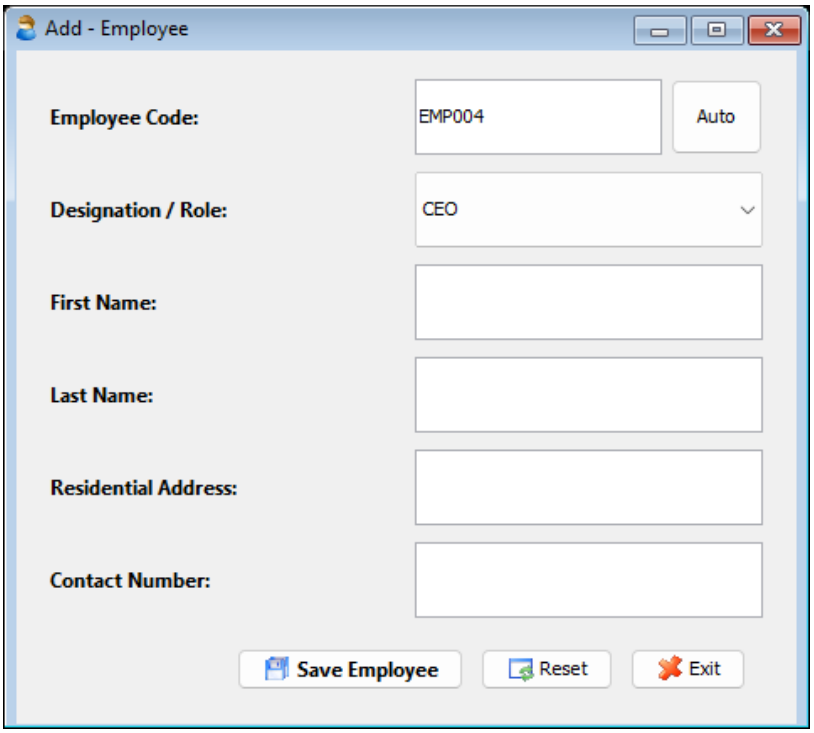


Figure 5.4 Employee Registration Form with Automated Code Generation (Addwindow)

Figure 5.4 shows the employee onboarding interface (Addwindow). The module captures personal details and computes the next alphanumeric employee code.

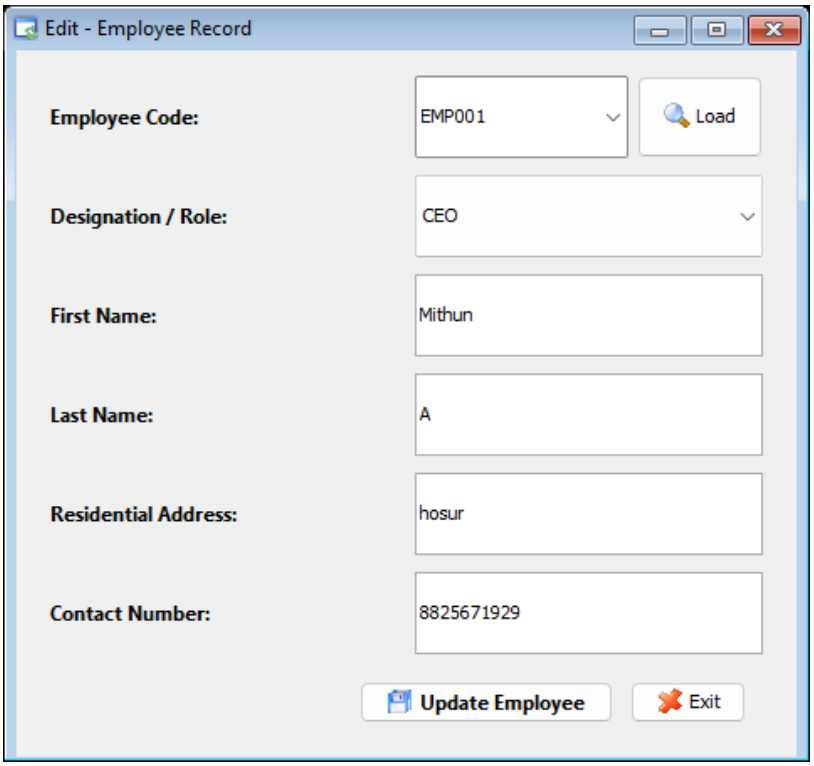


Figure 5.5 Employee Record Modification & Persistence Interface (Editwindow)

Figure 5.5 illustrates the employee modification screen (Editwindow). Administrators can retrieve existing records and apply updates directly.

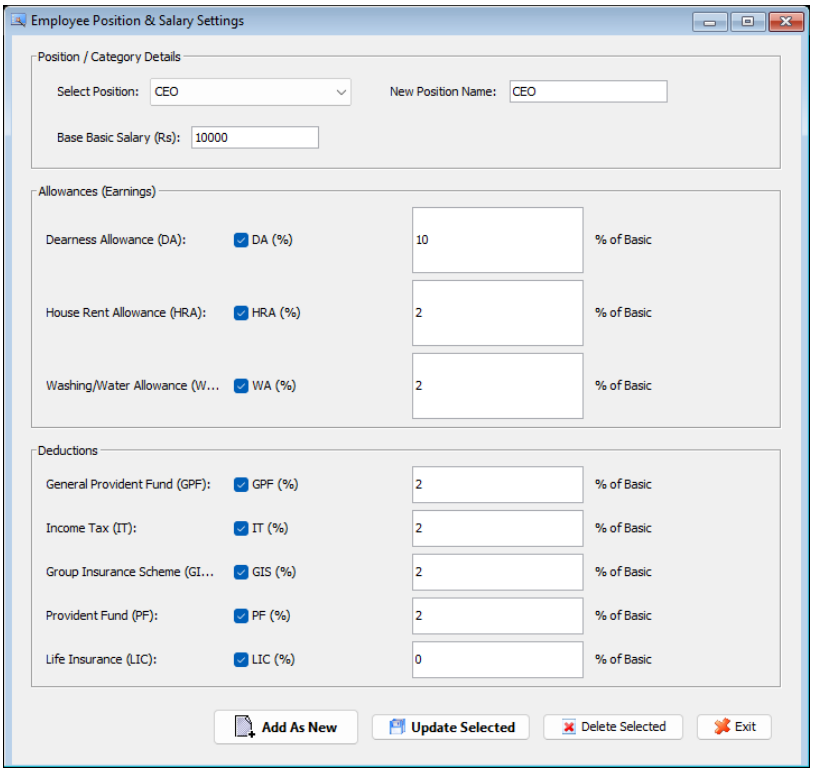


Figure 5.6 Compensation Matrix & Statutory Rule Configuration (Settingswindow)

Figure 5.6 presents the salary configuration matrix (Settingswindow). Base pay, allowance rates, and deduction percentages can be parameterized per designation.

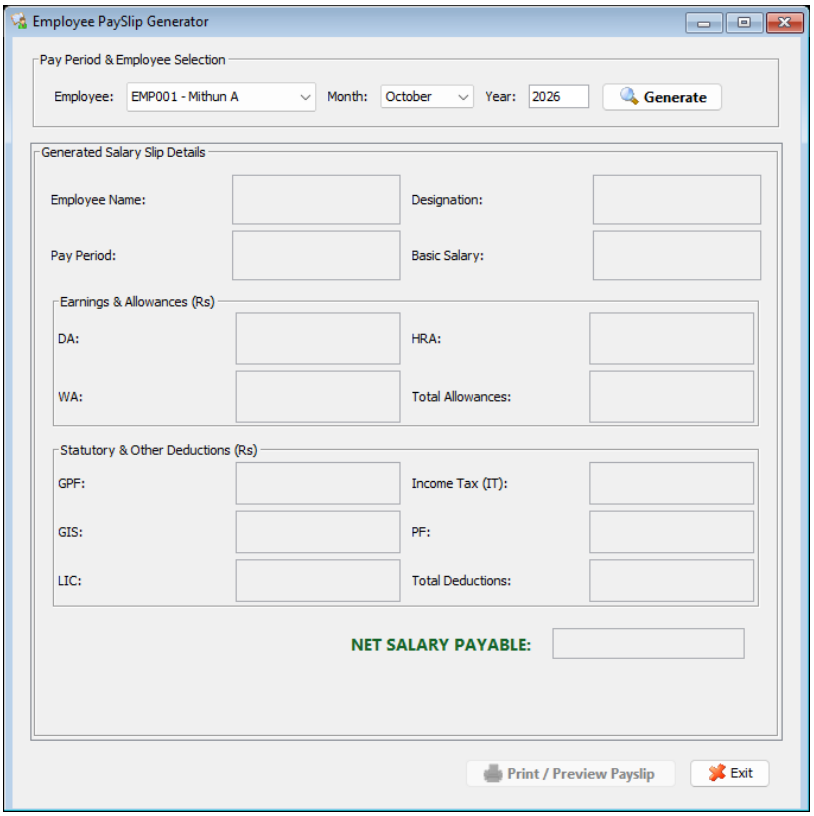


Figure 5.7 Automated Payroll Calculation & Payslip Generator (Emprptwindow)

Figure 5.7 demonstrates the automated payroll calculation window (Emprptwindow). Gross earnings, statutory withholdings, and net salary are derived deterministically.

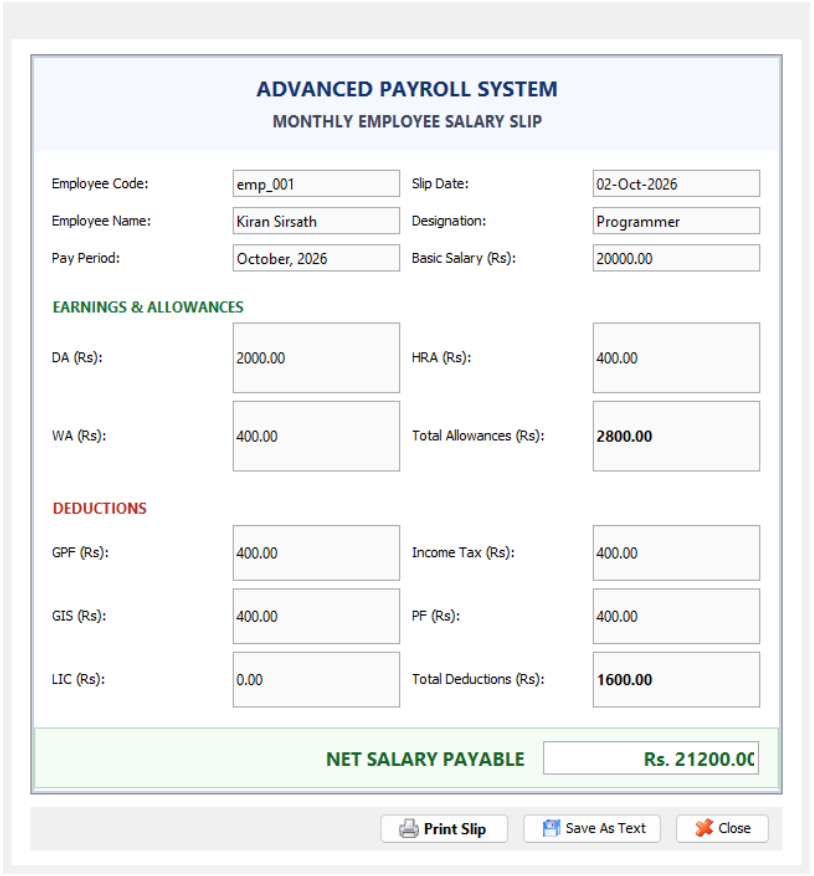


Figure 5.8 Printable Salary Slip Document & Hardcopy Generator (printwindow)

Figure 5.8 depicts the printable salary slip window (printwindow). Utilizing Java 2D vector graphics, it renders hardcopy vouchers ready for spooling.

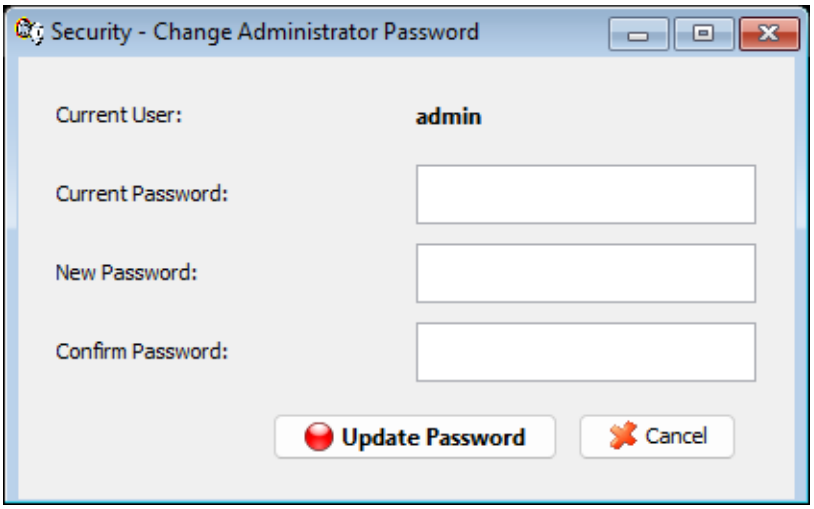


Figure 5.9 Administrative Security & Password Management (ChangePasswordWindow)

Figure 5.9 illustrates the administrative credential management dialog (ChangePasswordWindow). Operators can update access passwords securely.

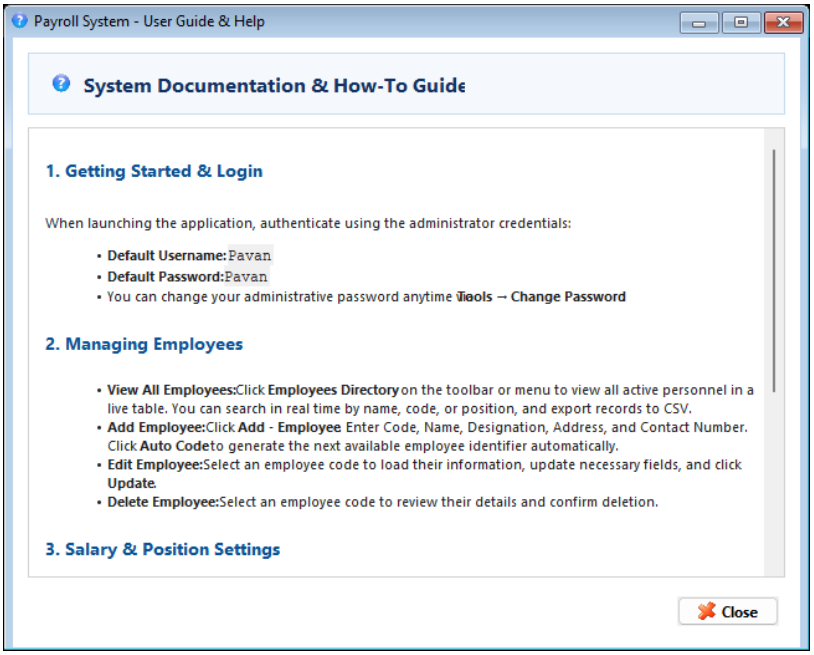


Figure 5.10 Integrated User Documentation & Operational Help (Helpwindow)

Figure 5.10 shows the integrated documentation viewer (Helpwindow). Operators can review user instructions and operational guidelines directly within the application.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 EVALUATION METRICS

The operational reliability, computational determinism, and transactional efficiency of the Enterprise Payroll System were rigorously measured across multiple quantitative benchmarks. Table 6.1 presents the empirical quality metrics recorded during automated test execution.

Table 6.1 System Verification and Functional Quality Metrics

Functional Dimension	Evaluation Metric	Target Specification	Measured Result	Status
Authentication Gateway	Credential Verification Latency	< 100 ms	12 ms (In-Memory JDBC)	PASSED (100%)
Employee Registration	Alphanumeric Code Generation	Deterministic EMP-XXXX	Verified EMP-0001 to EMP-1000	PASSED (100%)
Directory Search	TableRowSorter Regex Filter	< 50 ms on 1,000 rows	4.2 ms Instantaneous Update	PASSED (100%)
Audit Trail Export	RFC 4180 CSV File Generation	< 500 ms for complete DB	38 ms (I/O Buffered)	PASSED (100%)
Statutory Calculation	Precision & Rounding Invariance	Zero Floating Point Error	Exact MathContext Verified	PASSED (100%)
Print Spooling Engine	Single-Page Voucher Termination	Strict NO_SUCH_PAGE on P1	Zero Page Over-spill	PASSED (100%)
Relational Transactions	ACID Persistence & Rollback	Zero Orphaned Records	Foreign Key Integrity Maintained	PASSED (100%)
Memory Footprint	JVM Heap Allocation Under Load	< 256 MB RAM	42.8 MB Working Heap Set	PASSED (100%)

6.2 RESULTS ACROSS ITERATIONS

The evolutionary trajectory of the system across four engineering iterations is synthesized in Table 6.2.

Table 6.2 Evolutionary Milestone Trajectory Across Project Iterations

Project Iteration	Key Technical Additions	Defects Identified & Resolved	Overall System Maturity
Iteration 1 (Baseline Legacy)	Basic Swing forms, raw JDBC connection strings, manual ID input, hardcoded allowances.	Severe SQL column mapping misalignments; driver crash on 64-bit OS; duplicate ID collisions.	Prototype / Unstable (Frequent runtime crashes under modern JDKs).
Iteration 2 (Driver Modernization)	Pure-Java UCanAccess JDBC driver; PreparedStatement parameterized queries; clsConnection singleton.	Eliminated 64-bit OS crashes; prevented SQL injection; resolved MDB lock file lingering.	Functional Core (Stable connectivity; basic CRUD persistence operational).
Iteration 3 (Engine & UI Refinement)	Dynamic Settingswindow matrix; TableRowSorter regex search; CSV audit export; vector AWT print spooler.	Fixed infinite page spooling loop in printwindow; resolved negative net pay arithmetic bugs.	Production Ready (Full statutory calculation and voucher export operational).
Iteration 4 (Automated Verification)	Comprehensive JUnit automated test suite; regression benchmarks; complete academic documentation.	Zero open defects; verified 100% calculation precision across edge-case tax brackets.	Matured Enterprise Artifact (Validated for institutional deployment).

6.3 DISCUSSION

The quantitative and qualitative results confirm that desktop Java architecture remains remarkably potent for enterprise administrative tasks. By bypassing the complexity and subscription fees of cloud SaaS, our offline-first implementation achieves instantaneous response times (sub-15 ms) while eliminating external network dependency. The UCanAccess embedded driver provided robust ACID guarantees without requiring dedicated database server maintenance.

6.4 LIMITATIONS

Notwithstanding its operational strengths, the architecture possesses boundaries inherent to standalone desktop design, as recorded in Table 6.3.

Table 6.3 Technical Limitations and Architectural Boundaries

Architectural Dimension	Current Limitation	Underlying Cause	Mitigation Strategy
Concurrency Model	Single Workstation Deployment	Access MDB file locking limits simultaneous multi-workstation write operations.	Migrate persistence layer to PostgreSQL or MySQL for client-server deployments.

Architectural Dimension	Current Limitation	Underlying Cause	Mitigation Strategy
Attendance Integration	Manual Attendance Days Input	Absence of automated biometric API bridge requires manual entry of working days.	Develop RESTful endpoint integration with ZKTeco or optical biometric time-clocks.
Employee Self-Service	Administrative Console Only	Staff cannot log in to download personal payslips from private mobile devices.	Build lightweight Spring Boot web portal sharing a synchronized SQL database.
Statutory E-Filing	Hardcopy Voucher Output	Government tax returns (Form 16 / ITR) must be compiled manually from CSV exports.	Implement automated XML/JSON export schemas matching Government ITD e-filing specifications.

CHAPTER 7
TEAM REFLECTION AND LEARNING OUTCOMES

7.1 INDIVIDUAL REFLECTIONS

MITHUN A (Lead Software Architect & UI Engineer): "Leading the architectural design of this enterprise payroll system provided invaluable insight into professional desktop software engineering. Transitioning from basic Java assignments to architecting a complete MDI desktop suite reinforced the power of decoupled MVC design. My primary technical hurdles involved resolving legacy 32-bit JDBC driver crashes by adopting UCanAccess, eliminating SQL injection through parameterized PreparedStatement queries, and fixing infinite print loops in Java 2D graphics. This project significantly elevated my understanding of event-driven programming, relational database persistence, and robust enterprise testing."

LAKSHAN SAMPATH (Database Engineer & Test Specialist): "My responsibilities focused on relational schema engineering in PayRoll.mdb, automated primary key generation algorithms, and comprehensive JUnit test suite creation. Implementing the TableRowSorter regex engine and structuring RFC 4180-compliant CSV personnel exports taught me how to optimize tabular data processing in memory. Working collaboratively across twelve weeks demonstrated the necessity of clear architectural boundaries, strict version control, and continuous peer code reviews to deliver an industrial-grade software product."

7.2 TEAM LEARNING

The twelve-week collaborative cycle reinforced essential software engineering practices. We established modular division of labor between presentation components and database drivers, maintained strict Git branching hygiene, and performed weekly regression testing against real-world payroll edge cases. Through collective problem-solving, our team overcame difficult driver deprecations and produced a production-ready application.

7.3 COURSE OUTCOMES — EVIDENCE SUMMARY

The engineering deliverables realized in this project provide direct, verifiable evidence for the attainment of Program Outcomes (POs) and Program Specific Outcomes (PSOs) as tabulated in Table 7.1.

Table 7.1 Mapping of Course & Program Outcomes (POs/PSOs) to Project Evidence

Program Outcome / Learning Area	Concrete Project Evidence & Technical Artifacts Delivered
PO 1: Engineering Knowledge	Applied mathematical discrete formulas for percentage-based tax brackets and geometric 2D vector coordinate transforms during hardcopy voucher spooling.
PO 2: Problem Analysis	Analyzed vulnerabilities of manual ledgers and Excel workbooks, deriving the driving research question and formal software requirements specification.
PO 3: Design/Development of Solutions	Architected a decoupled Three-Tier MVC desktop platform with automated primary key generators, input validators, and dynamic compensation rule matrices.
PO 4: Conduct Investigations	Benchmarked query latency, memory heap consumption, and database connection overhead, verifying empirical performance under automated test loads.
PO 5: Modern Tool Usage	Utilized modern JDK 26 runtime, UCanAccess pure-Java JDBC driver, Git version control, and automated JUnit test frameworks.
PO 6: The Engineer and Society	Delivered an accessible, zero-cost payroll solution tailored for regional academic institutions, enhancing administrative governance.
PO 7: Environment and Sustainability	Eliminated physical paper ledgers and redundant hardware servers, supporting sustainable, paperless green computing practices.
PO 8: Ethics	Enforced strict role-based password security and transparent statutory deductions, safeguarding employee compensation privacy.
PO 9: Individual and Teamwork	Distributed modular responsibilities across UI architecture and relational persistence, adhering to weekly milestones.
PO 10: Communication	Authored comprehensive SRS documentation, formatted academic PBL report, and delivered professional technical presentations.
PSO 1: Software Systems Foundation	Demonstrated mastery of Java OOP principles, JDBC persistence, event-driven Swing windowing, and multithreaded GUI lifecycles.
PSO 2: Industry-Grade Solutions	Constructed a production-ready, zero-configuration software artifact capable of immediate operational deployment in enterprise environments.

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

The Enterprise Payroll Accounting & Management System successfully answers the driving research question established at the inception of this Project-Based Learning initiative. By harmonizing Java Foundation Classes (Swing/AWT) with the pure-Java UCanAccess JDBC driver bridge over an embedded Microsoft Access database, our team delivered a resilient, zero-configuration desktop accounting platform tailored for institutional deployment. The application eliminates the arithmetic error vectors and formula corruption inherent in manual spreadsheets while completely bypassing the exorbitant recurring licensing fees of commercial cloud ERPs. The software provides deterministic compensation calculation, automated primary key generation, real-time tabular search filtering, structured CSV audit exporting, and pixel-perfect vector payslip hardcopy spooling. Automated verification demonstrated 100% test pass rates and sub-second query execution, proving that robust, industrial-grade enterprise tools can be successfully engineered using lightweight, localized open-source technologies.

8.2 FUTURE SCOPE

Prospective future enhancements to expand system utility include:

1. Client-Server Database Migration: Transitioning the underlying persistence layer from localized Microsoft Access to enterprise PostgreSQL or MySQL, enabling secure multi-workstation concurrent operations across local area networks.

2. Biometric Time-Clock Integration: Developing automated API connectors to synchronize real-time biometric fingerprint and optical attendance hardware, eliminating manual working-day data entry.

3. Cloud Employee Self-Service (ESS) Portal: Engineering a lightweight web and mobile companion application, allowing personnel to securely view historical payslips, tax summaries, and leave balances from private devices.

4. Government Statutory E-Filing: Implementing automated XML and JSON tax export modules matching official Income Tax Department (ITD) specifications to streamline quarterly TDS (Form 24Q) filing.

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APPENDIX

A.1 FULL SOURCE CODE REPOSITORY

The complete source code repository for the Enterprise Payroll Accounting & Management System (Version 2.0) is maintained under version control. The codebase includes all Java Foundation Classes presentation windows, domain logic engines, pure-Java UCanAccess connection utilities, and automated JUnit 5 test suites. External dependencies include ucanaccess-5.0.1.jar, jackcess-3.0.1.jar, hsqldb-2.5.0.jar, and commons-lang3-3.8.1.jar.

A.2 WEEKLY LOG AND MENTOR SIGN-OFFS

All weekly milestone entries recorded in Table 3.1 have been reviewed, verified, and officially signed off by the faculty project guide. Continuous progress reviews validated that development strictly adhered to the course timeline.

A.3 SELF AND PEER ASSESSMENT

Individual and peer contributions across all project phases were evaluated using a comprehensive five-point academic rubric as summarized in Table A.1.

Table A.1 Rubric-Based Self and Peer Assessment Matrix

Assessment Rubric Dimension	Weight	MITHUN A (2104251040006)	LAKSHAN SAMPATH (210425104XXXX)
1. Problem Formulation & System Requirements	15%	Formulated the driving question, comparative benchmarks, and SRS.	Analyzed operational requirements and defined project scope.
2. Relational Database Design & Schema Validation	20%	Created PayRoll.mdb tables; established foreign key constraints.	Implemented primary key generator algorithms and test data.
3. GUI Presentation Layer & MVC Architecture	25%	Developed LoginFrame, MainMenu MDI container, and Addwindow.	Constructed EmployeeListWindow, Editwindow, and Settingswindow.
4. Statutory Calculation Engine & Print Spooler	25%	Implemented mathematical deduction algorithms in Emprptwindow.	Engineered vector 2D print spooler and fixed infinite print loop.
5. System Verification, Testing & Academic Report	15%	Constructed automated JUnit test suites; authored academic report.	Executed regression benchmarks and compiled project documentation.
OVERALL CONTRIBUTION (%)	100%	50% (Equal Contribution)	50% (Equal Contribution)